Supplementary material for

What ParallaxBA Actually Buys: Better Scaling, Not More Information

This supplement collects three tables that support identities and statements in the main text without carrying its argument. Section, equation and table numbers cited in the captions refer to the main text.

S1 Rank bookkeeping for an exactly collinear point

The configuration of Section 4.3: four cameras at $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, $(0,0,1)$, five generic points, and one point $X_5 = (2,0,0)$ on the ray through the first two centres, observed only by them.

Table S1: Rank bookkeeping for a BA problem with and without an exactly collinear point. The extra point adds one null direction to H but none to $\mathcal{S}$, as predicted by the rank identity of the main text.

Quantity	Base	With X_5	Change
Variables	39	42	+3
rank H	32	34	+2
dim ker H	7	8	+1
rank J_X	15	17	+2
rank $\mathcal{S}$	17	17	0
dim ker $\mathcal{S}$	7	7	0
rank H − rank J_X	17	17	0
Rank of the X_5 block	—	2	—

S2 Damped steps along depth under a shared schedule

The two-view configuration of Section 3.4 with $d/B \approx 50$ and a five-percent initial depth error. The two columns are the undamped depth steps the two charts take from the same physical state under the same damping parameter; the third column maps the chart step to physical coordinates through D_q.

Table S2: Single-point Levenberg–Marquardt steps along depth under the same λ schedule and the same physical initial state ($d/B \approx 50$, five-percent depth error). The third column maps the chart step to physical coordinates through D_q.

λ	XYZ depth step	Parallax depth step	Relative gap
10^{-12}	-2.62456	-2.62460	1.5×10^{-5}
10^{-10}	-2.62062	-2.62460	1.5×10^{-3}
10^{-8}	-2.27848	-2.62460	0.152
10^{-6}	-0.16231	-2.62459	15.0
10^{-4}	-1.94×10^{-3}	-2.62408	119
10^{-2}	-3.34×10^{-5}	-2.57340	1522
1	-3.45×10^{-7}	-1.05003	5.8×10^{4}

S3 Residuals of the exact identities

The five-camera, fifteen-point problem of Section 4.6 with seventy-five observations: 150 residuals, 75 variables, anchors 0 and 1.

Table S3: Residuals of the exact identities of Section 4 on a generic five-camera BA problem, except where a single-point reduction is indicated.

Check	Maximum discrepancy
Chart reconstruction $X = \Phi_\xi(\Phi_\xi^{-1}(X))$	2.66×10^{-15}
Analytic chart Jacobian versus central differences	1.62×10^{-8}
$H_q - T^\mathsf{T} H_x T$	1.78×10^{-15}
$\mathcal{S}_q - \mathcal{S}_x$	7.11×10^{-15} (relative 5.95×10^{-16})
$g_q - T^\mathsf{T} g_x$	2.22×10^{-16}
Reduced gradients, both charts	2.69×10^{-15}
Damping identity, full BA	1.67×10^{-13}
Damping identity, single point	4.42×10^{-12}